10 A shop sells

17 different types of sandwich
15 different types of drink
26 different types of snack
14 different types of dessert.

A meal deal consists of either

a sandwich, a drink and a snack
or a sandwich, a drink and a dessert.

(a) Show that there are over 10 000 different ways to choose a meal deal.

Sandwich, drink, snack: $17 \times 15 \times 26 = 6630$

1 mark
Multiply independent choices.

Sandwich, drink, dessert: $17 \times 15 \times 14 = 3570$

Total ways = $6630 + 3570 = 10200$

1 mark
Add the two alternatives.

$10200 > 10000$, as required.

1 mark
10200.

(3)

The owner of the shop says,

> "If I halve the number of snacks available, this will halve the number of ways to choose a meal deal."

The owner of the shop is incorrect.

(b) Explain why.

Only snack choices halve; dessert choices stay the same.

1 mark
One option set changes.

New total = $17 \times 15 \times (13 + 14) = 6885$,

which is more than $10200 \div 2 = 5100$.

(1)

(Total for Question 10 is 4 marks)

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